BE sem VII | IT | (-scheme) Nov 24 | 14.11.25
qp code - 95963
(1/1)

Time (3 Hours) **[Total Marks 80]**

Note: 1. Question 1 is compulsory
2. Answer any three out of remaining questions
3. Assume suitable data where required

Q1 Solve any 4

a) Define smart object and its Characteristics in brief. 5
b) Define the following i) Fog Computing ii) Internet of Behaviour (IoB). 5
c) Differentiate briefly Physical design of IoT and Logical design of IoT. 5
d) Explain the importance of IoT Data Analytics. 5
e) Write a short note on Near Filed Communication. 5

Q2

a) Explain the IOT enabled Smart Parking system with respect to sensors, Actuators, framework, protocols, data analysis and security. 10
b) Explain the following terms 10
i) Constrained Application Protocol
ii) Streaming Text Oriented Message Protocol

Q3

a) Elaborate briefly the simplified 3 layered IoT architecture with neat diagram, 10
b) Describe the role of NoSQL in IoT Data Analytics Challenges. 10

Q4

a) Explain effective Data Visualization methods With suitable example 10
b) Explain in brief the MQTT Architecture with Advantages. 10

Q5

a) State the Comparison's of all Protocols used in IOT for cloud Edge. 10
b) Describe the Architecture of Advanced Message Queuing Protocol (AMQP) with major application. 10

Q6

a) Explain the architecture of RFID with its major applications. 10
b) Illustrate the components of IEEE 802.11 architecture with advantages 10

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04/06/2025 BE IT SEM-VII C-SCHEME IOE QP CODE: 10083969

[3 hrs] [80 Marks]

Note : 1. Question 1 is compulsory
2. Answer any three out of remaining questions
3. Assume suitable data where required

Q1	Solve any four	
a)	Define Smart Object in IoT and its characteristics.	**5 marks**
b)	Write short note on RFID	**5 marks**
c)	Draw the Architecture diagram of Advanced Message Queuing Protocol (AMQP) with advantages	**5 marks**
d)	Explain the importance of IoT Data Analytics.	**5 marks**
e)	Explain data acquisition in IoT.	**5 marks**
Q2		
a)	Describe IoT World Forum (IoTWF) standardized architecture.	**10 marks**
b)	State various operating modes of Zigbee and its topologies with diagrams.	**10 marks**
Q3		
a)	Draw and explain the architecture of MQTT with diagram.	**10 marks**
b)	Explain working of Constraint Application Layer protocol (CoAP).	**10 marks**
Q4		
a)	Illustrate the following i) How to create and visualize alerts in IoT? ii) What are the main challenges of IoT?	**10 marks**
b)	Explain the following terms: i) CoAP ii) Internet of Behaviour	**10 marks**
Q5		
a)	Explain the architecture of LoRaWAN with its major characteristics.	**10 marks**
b)	Design the Forest Fire Detection system using IoT sensors.	**10 marks**
Q6		
a)	Describe IoT enabled Smart Home Automation.	**10 marks**
b)	Explain the role of NoSQL in IoT data analytics.	**10 marks**

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[3 hrs] [80Marks]

Note: 1. Question 1 is compulsory
2. Answer any three out of remaining questions
3. Assume suitable data where required

Q1	Solve any 4	
a)	Explain functional blocks of IoT.	5
b)	Explain the importance of IoT Data Analytics.	5
c)	Write a short note on Near Filed Communication.	5
d)	Define the following i) Fog Computing ii) Internet of Behaviour (IoB).	5
e)	Differentiate briefly Physical design of IoT and Logical design of IoT.	5
Q2		
a)	Explain the IOT enabled Home Automation system with respect to sensors, Actuators, framework, protocols, data analysis and security.	10
b)	Explain the following terms i) CoAP ii) Data Lakes.	10
Q3		
a)	Explain the architecture of LoRaWAN with its major Characteristics	10
b)	Illustrate the components of IEEE 802.11 architecture with advantages	10
Q4		
a)	Explain effective Data Visualization methods With suitable example	10
b)	Explain in brief the MQTT Architecture with Advantages.	10
Q5		
a)	State the Comparison's of all Protocols used in IOT for cloud Edge.	10
b)	Describe the Architecture of Advanced Message Queuing Protocol (AMQP) with major application.	10
Q6		
a)	Elaborate briefly the simplified 3 layered IoT architecture with neat diagram,	10
b)	Describe the role of NoSQL in IoT Data Analytics Challenges.	10

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Time: 3 hrs Marks: 80

Note: 1. Question 1 is compulsory
2. Answer any three out of remaining questions
3. Assume suitable data where required

Q1	Solve any 4	
a)	Explain Bluetooth Low Energy(BLE) role	5
b)	Briefly elaborate the COAP	5
c)	Explain data retention strategy.	5
d)	Explain the concept of I-IoT and its similarity with IoT	5
e)	Explain the characteristic of IoT	5
Q2		
a)	How can IoT analytics be effectively utilized within IoT-based healthcare systems? Additionally, what are some essential parameters that should be incorporated into the patient dashboard for comprehensive monitoring and management of health data?	10
b)	Evaluate long-range communication systems and protocols such as LTE, LTE-A, LoRa, and LoRaWAN in the context of IoT connectivity. Discuss their suitability for different IoT use cases based on factors like coverage, data rate, power consumption, and scalability.	10
Q3		
a)	Define the role of analytics in IoT technology and elaborate the challenges associated with it.	10
b)	Elaborate the need of new network architecture in IoT.	10
Q4		
a)	Compare edge, fog and cloud computing w.r.to its hierarchy.	10
b)	Consider smart smoke detection system. Elaborate its working and list down the different types of sensors and actuators required during the deployment scenario.	10
Q5		
a)	Explain the role of HTTP, WebSocket, and MQTT in IoT communication. Compare and contrast these protocols in terms of their characteristics, suitability for different IoT scenarios, and support for real-time data transmission.	10
b)	Discuss the functional blocks of IoT architecture, highlighting their roles and interactions. Provide examples to illustrate the importance of each block in the overall functionality of IoT systems.	10
Q6		
a)	Elaborate the Smart Object with diagram and describe its characteristics.	10
b)	Explain the following access technologies with applications area of each 1) IEEE 802.15.4 2) Z-wave 3) LTE-A	10

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28-Dec-2023 10:30 am - 01:30 pm 1T01237 - B.E.(Information Technology Engineering) (SEM-VII)(Choice Base Credit Grading System) (R- 19) (C Scheme) / 42672 - Internet of Everything QP CODE: 10038114

Time : [3 hrs] **[80 Marks]**

Note : 1. Question 1 is compulsory

2. Answer any three out of remaining questions

3. Assume suitable data where required

Q1	Solve any 4		
a)	Explain ZigBee in brief.		5
b)	Explain the Need of lightweight new communications protocols for IoT.		5
c)	Explain data retention strategy.		5
d)	Write a short note on I-IoT		5
e)	Explain functional blocks of IoT		5
Q2 a)	Elaborate in detail the strategies to organize Data for IoT Analytics.		10
b)	Illustrate the role of Data refineries in preventing data lakes to turn into data swamps		10
Q3 a)	Describe the communication system and protocols involved in long range communication system of IoT		10
b)	With neat diagram, elaborate briefly the simplified 3 layered IoT architecture.		10
Q4 a)	Differentiate and give Hierarchy between Edge, Fog, and Cloud computing		10
b)	Consider Smart Irrigation system. Elaborate its working with block diagram and list down the different types of sensors and actuators required during the deployment scenario.		10
Q5 a)	Elaborate MQTT with its working in details and two advantages over COAP.		10
b)	Illustrate 3 methods for effective Data Visualization		10
Q6 a)	Define sensor, actuators and their role and classifications in brief.		10
b)	Explain the following access technologies with applications area of each 1) IEEE 802.15.4 2) RFID		10

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1T01237 - B.E.(Information Technology Engineering)(SEM-VII)(Choice Base Credit Grading System) (R- 19) (C Scheme) / 42672 - Internet of Everything
QP CODE: 10026510 DATE: 16/06/2023

[3 hrs] **[80 Marks]**

Note : 1. Question 1 is compulsory.
2. Answer any three out of remaining questions.
3. Assume suitable data where required.

Q1 Solve any 4

a)	What is the Internet of Behaviour (IoB) and why is it the future?	5
b)	Compare and contrast IoT data analytics and IoT network analytics.	5
c)	Briefly elaborate the Advanced Message Queuing Protocol(AMQP)	5
d)	While designing the IoT based smoke detection system what all parameters need to be considered before choosing the required senor.	5
e)	Explain applications of Bluetooth Low Energy (BLE)	5

Q2

a)	Compare and contrast Edge, Fog, and Cloud Computing. Explain their usage with an example.	10
b)	Consider smart farming system. Elaborate its working and list down the different types of sensors and actuators required during the deployment scenario	10

Q3

a)	Describe IoT World Forum (IoTWF) Standardized Architecture	10
b)	Explain the architecture of LoRaWAN with its major Characteristics	10

Q4

a)	Explain ecosystem for IoT enabled Smart Home Automation with respect to sensors, actuators, framework, protocols, storage, data analysis, security etc.	10
b)	State advantages of data visualisation. Give different tools used for it.	10

Q5

a)	Elaborate the Smart Object with diagram and describe all the essential components required to make the object into the smart object.	10
b)	Explain the following access technologies with applications area of each 1) Zigbee 2)RFID 3)6LoWPAN 4) CoAP	10

Q6

a)	Draw and explain Architecture of MQTT with diagram	10
b)	Elaborate the need of new network architecture in IoT.	10

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